

SIT281 Cryptography - PBL A (Week 4)

1. Select one pair of numbers from Set A and Set B.

Set A : (2050, 371), (3300,255), (3205,495)

Set B: (4267,310), (3450,317), (4205,332)

- a. Factorize the two numbers in each pair.
 - b. Find the GCD and LCM of the pairs using the factors.
 - c. Find the GCD of the pairs using EA.
 - d. Find the inverse (of the smaller number in modulo of the bigger number) using EEA. What is the condition to be satisfied by the two numbers for the existence of an inverse ?
2.
 - a. Convert the decimal number (Select one number from 4572, 4275, 4177, 4087) into Binary. Using the Binary code, convert it to Octal and Hexadecimal
 - b. Verify FLT using an example. Using FLT, verify that 97 is Prime.
 - c. Calculate $\Phi(15)$, $\Phi(11)$ and $\Phi(10)$ and verify the result by writing and counting the corresponding relative prime numbers of each of the cases.
 - d.
 - i. Calculate $3^{2451} \bmod 97$ using FLT
 - ii. Calculate $3^{2451} \bmod 120$ using Euler Theorem
 - e. Solve the following two equations using CRT
 $x = 4 \bmod 5$ and $x = 1 \bmod 7$
 3.
 - a. Encrypt a letter of your choice (A (0) to Z(25)) the Affine Cipher $C = 7P + 6 \bmod 26$. Verify decrypted results. Show the calculation of finding the multiplicative inverse. Can $C = 13P+6$ be used as Encryption equation ? Why ?
 - b. What is the difference between mono- alphabetic ciphers and poly-alphabetic ciphers ? Which one is vulnerable for Frequency Analysis ? Briefly describe Frequency Analysis technique.
 - c. Using a key 3, 5 to encrypt a letter (twice) of your choice, show that Vignere Cipher is polyalphabetic
 - d. Using the same letter (4 times) of your choice in a 2×2 matrix, and a matrix key $\begin{bmatrix} 5 & 3 \\ 4 & 4 \end{bmatrix}$, demonstrate the poly-alphabetic nature of Hill Cipher. Find the Inverse matrix using SageMath, and verify the decrypted result. Show that the key and the inverse key produce an identity matrix when multiplied together.